Khushi Joshi
Professor William Crowder
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Modelling Large Open Economies With International Linkages: Replicating Dungey and Osborne (2014) for the US and Euro Area.

1.0 Problem Statement and Motivation

The goal of this project is to replicate the key empirical findings of Dungey and Osborn (2014) on the economic interdependence between the United States and the Euro Area. Because these two major economies are deeply connected, I aim to understand how sudden economic disturbances, or "shocks" (such as unexpected changes in inflation or interest rates), spread, or "spill over," from one region to the other.

Dungey and Osborn used a Structural Vector Error Correction Model (SVECM) to analyze these linkages. This technique is specialized for separating different causes of change (like supply shifts versus monetary policy actions) and tracking their effects over time using Impulse Response Functions (IRFs). The original study concluded that U.S. monetary policy significantly influences the Euro Area, primarily transmitted through the exchange rate.

By recreating their analysis, this project seeks to confirm these crucial conclusions regarding the nature and asymmetry of international spillovers between the two largest open economies.

1.1 Economic Problem and Research Question

The economic problem addressed by Dungey and Osborn (2014) concerns the transmission of macroeconomic shocks between large open economies. Specifically, the paper asks:

- How do U.S. output, inflation, monetary policy, and exchange rate shocks affect the Euro Area?
- How do Euro Area shocks feed back into the U.S. economy?
- What role does the real exchange rate play in transmitting international disturbances?

Understanding these questions is important for evaluating the degree of international interdependence, the limits of monetary policy autonomy, and the strength of real and financial linkages between major economies.

1.2 Theoretical Background

- Open-Economy Macroeconomics: Large open economies influence each other through trade flows, capital markets, and expectations. Mechanisms such as interest rate parity, purchasing power parity, and exchange rate adjustments imply that shocks in one country can quickly spread to others.
- Structural Vector Autoregressions (SVARs): SVAR models allow researchers to identify structural shocks and trace their dynamic effects. Identification relies on restrictions based on economic theory, such as assumptions about which shocks have permanent effects or which variables adjust slowly. The original study uses both short-run and long-run restrictions to distinguish domestic and international shocks.

- Cointegration and Long-Run Equilibrium: Macroeconomic variables like output, prices, and interest rates often share long-run relationships. A Vector Error Correction Model captures these shared trends while allowing for short-run deviations. In the Dungey–Osborn framework, long-run restrictions reflect theoretical assumptions that only supply shocks have permanent effects on output, while demand-side and monetary shocks are temporary.

2.0 Data Description

This study uses quarterly macroeconomic data for the United States and the Euro Area from 1983:Q1 to 2007:Q4, matching the sample period of Dungey and Osborn (2014). The dataset includes the key variables studied in the original model: real output, consumer price inflation, short-term interest rates, the nominal exchange rate, the real exchange rate, and oil price inflation. Data fidelity was maintained by sourcing US data from the FRED database, while Euro Area output data utilized the Area Wide Model (AWM) database, and inflation and interest rate data incorporated the historical series constructed by Anderson et al. (2011).

Real output for both regions is converted to natural logarithms to allow interpretation in terms of growth rates. Inflation is derived from consumer price indices and converted from annualized percentage rates into quarterly log changes by dividing by 400. Log price levels are then constructed by cumulatively summing these quarterly increments, with each series initialized at ln(100) to replicate the original authors' scaling. The bilateral real exchange rate is calculated as the log nominal exchange rate plus the difference between the constructed log price levels of the Euro Area and the United States. This ensures consistency with open-economy macroeconomic theory, where real exchange rate movements reflect relative price changes.

3.0 Econometric Methodology

3.1 Model Specification and Replication Challenges

The data construction and lag-length selection closely match the design of Dungey and Osborne (2014). All seven variables were transformed according to the original study, and the lag length p=3 in levels was retained, ensuring a high degree of consistency with the authors' baseline specification. However, while the data preparation aligns almost exactly with the original study, full replication of the structural results is not possible because their identification strategy cannot be reproduced using standard SVECM tools in R. Dungey and Osborne relied on a uniquely tailored Structural VECM, built through a manual, equation-by-equation identification process. This approach involved:

- Cointegration-based long-run restrictions: Three variables (US output, EA output, and the real exchange rate) were treated as I(1) and linked by a single cointegrating relationship, allowing the authors to distinguish permanent from temporary shocks in the long run.
- Custom short-run restrictions: Rather than using automated SVECM routines, the authors imposed contemporaneous causal structure by hand using OLS, embedding assumptions such as US leadership in real activity and inflation, and monetary policy reacting mainly to domestic conditions.
- Because this manual structural identification cannot be replicated exactly in R, I adopt a feasible alternative: a standard VAR(3) in levels identified using a recursive Cholesky decomposition. Although this approach does not recover the same theoretically grounded structural shocks as the SVECM, it does allow me to replicate the broad dynamic

behavior of the system. The resulting orthogonalized impulse response functions (O-IRFs) produce interpretable responses of each variable to shocks in the others and preserve key qualitative patterns—such as the direction, timing, and short-run transmission of macroeconomic shocks.

While the replication does not reproduce the exact structural decomposition of the original SVECM, it successfully replicates the overall dynamic relationships in a statistically coherent way, using an alternative and implementable identification method.

3.2 Time Series Properties and Cointegration

The empirical analysis begins by establishing the time-series properties of the seven macroeconomic variables (Y^t). Since macroeconomic time series often exhibit non-stationarity, the analysis first confirms that all variables are integrated of order one, I(1) (stationary only after first differencing). This is a prerequisite for utilizing the Vector Error Correction Model (VECM), which is designed to handle non-stationary, but cointegrated, data. The VECM framework relies on the existence of cointegration, meaning the non-stationary variables share a long-run equilibrium relationship. To determine the number of these relationships, the Johansen Trace Test was employed, using a lag length (p=3) suggested by the underlying Vector Autoregression (VAR) estimation.

Table 1: Johansen Trace Test for Cointegration Rank (r)

Hypothesis	Test Statistic	10% Critical Value	5% Critical Value	1% Critical Value
r=0	199.54	126.58	131.7	143.09
r≤1	134.48	97.18	102.14	111.01
r≤2	75.95	71.86	76.07	84.45
r≤3	43.55	49.65	53.12	60.16
r≤4	23.77	32	34.91	41.07
r≤5	10.31	17.85	19.96	24.6
r≤6	2.42	7.52	9.24	12.97

The test results presented in Table 1 strongly reject the null hypothesis of no cointegration (r=0) and one cointegrating vector (r≤1) at the 1% significance level. The null hypothesis of r≤2 is rejected at the 10% level but not at the 5% level. Following the original Dungey and Osborne (2014) paper, which argues for a single, theoretically motivated long-run relationship derived from economic theory, a cointegration rank of r=1 is selected for the primary analysis.

3.3 The Vector Error Correction Model (VECM)

The VECM for the vector of variables is as follows:

$$\mathbf{Y}_t = [y_{\text{us},t}, y_{\text{eu},t}, \pi_{\text{us},t}, \pi_{\text{eu},t}, r_{\text{us},t}, r_{\text{eu},t}, \ln \text{RER}_t]'$$

Furthermore, the model integrates the long-run equilibrium (captured by $\beta' Y_{t-1}$) with the short-run dynamics (captured by the lagged difference terms, $\Gamma_i \Delta Y_{t-i}$).

$$\Delta \mathbf{Y}_t = \boldsymbol{\alpha}\boldsymbol{\beta}' \mathbf{Y}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i \Delta \mathbf{Y}_{t-i} + \mathbf{u}_t \quad \text{with } p = 3$$

3.3.1 Long-Run Cointegrating Vector and Adjustment

Assuming a cointegration rank of r=1, the VECM estimation yields a single, normalized cointegrating vector (β) and a matrix of adjustment coefficients (α). The estimated coefficients are presented in Table 2, where β defines the long-run relationship and α measures the speed at which each variable adjusts to deviations from this equilibrium.

Table 2: Cointegrating Vector (β) and Adjustment Coefficients (α)

Variable	β (Long-Run Vector)	α (Adjustment Coefficient)
y_{us}	**1**	0.0213 (0.0049)***
y_{eu}	-1.5482	-0.0069 (0.0048)
π_{us}	-0.1852	1.6214 (0.5643)**
π_{eu}	0.1428	-0.1269 (0.3678)
r_{us}	-0.0174	0.5397 (0.4346)
r_{eu}	-0.0138	-0.1547 (0.3764)
lnRER	0.5163	0.1422 (0.0472)**

Significance Codes: ***: 1%, **: 5%, *: 10%, .: 15%

- Interpretation of (β): The long-run relationship, normalized on y_{us}, shows a positive link between US output and the real exchange rate, and a negative link with Euro Area output.
- Interpretation of (α): Key adjustment processes are observed in the equations for US output (y_{us}), US inflation (π_{us}), and the real exchange rate (lnRER), all of which exhibit statistically significant coefficients, suggesting these variables play a role in restoring equilibrium after a shock.

3.3.2 Short-Run Dynamics

The Γ_i matrices capture the remaining short-run dynamic dependencies among the variables. Table 3 summarizes the coefficients from the first and second lagged differences (p -1 = 2).

Table 3: Short-Run Coefficients (Γ_1 and Γ_2)

Equation	Lag1_yus	Lag2_yus	Lag1_yeu	Lag2_yeu	Lag1_πus	Lag2_πus	Lag1_πeu	Lag2_πeu	Lag1_rus	Lag2_rus	Lag1_reu	Lag2_reu	Lag1_lnRER	Lag2_lnRER
Δy_{us}	-0.0424	0.1509.	0.2598**	0.1154	0.0016.	0.0016.	-0.0016	0.0024	-4.00E-04	-3.00E-04	9.00E-04	9.00E-04	-0.0255**	-0.0248**
Δy_{eu}	0.1053	0.0772	0.085	-0.023	-0.0033***	0	0.0032.	7.00E-04	8.00E-04	0.0025*	-0.0024*	-0.0024*	0.022*	-0.006
$\Delta \pi_{us}$	-1.4067	3.5272	17.0578.	-8.5429	0.0623	0.0709	0.2327	-0.0643	0.1133	0.0545	0.2217.	0.2217.	-0.3313	0.0202
$\Delta \pi_{eu}$	4.5723	-12.1335.	10.0178	-6.7309	-0.0509	-0.0235	0.0284	-0.0408	-0.0129	0.1388.	0.1948*	0.1948*	-1.0679	-0.3256
Δr_{us}	13.1527.	-12.2363	7.668	10.6317	0.1092	0.0418	0.0928	-0.0284	0.4824***	0.0692	-0.0401	-0.0401	0.3066	0.3066
Δr_{eu}	-2.6388	-10.8554.	9.1111	25.97***	-0.0808	-0.045	0.0403	-0.1381	0.4224***	-0.0123	-0.0255	-0.0255	-0.1885	-0.1885
$\Delta lnRER$	-1.8362*	-0.5051	1.6701.	2.1462**	-0.0058	0.0026	0.0368*	0.0091	0.0039	-0.0226*	-0.001	-0.001	-0.0551	-0.0551

Significance Codes: ***: 1%, **: 5%, *: 10%, .: 15%

3.4 Structural Identification and Impulse Response Analysis

The core of my analysis is to interpret the model's results in terms of true economic causes, or "structural shocks." The basic forecast errors (residuals) from the VECM are just correlated noise (u_t). To find the meaningful, uncorrelated structural shocks (ε_t), these errors must be transformed using a matrix A: $u_t = A\varepsilon_t$.

Dungey and Osborn (2014) use a Structural VECM (SVECM) approach, which imposes specific theoretical restrictions on matrix A to cleanly identify different shock types (like supply, demand, and monetary policy for each region).

Before applying the full structural identification, I first calculated the Orthogonalized Impulse Response Functions (O-IRFs) using a simple Cholesky decomposition of the variance-covariance matrix. This decomposition forces a recursive ordering (in this case, $\Delta y_{us} \rightarrow \Delta y_{eu} \rightarrow \Delta\pi_{us} \rightarrow \Delta\pi_{eu} \rightarrow \Delta r_{us} \rightarrow \Delta r_{eu} \rightarrow \Delta lnRER$), meaning earlier variables instantaneously affect later ones, but not vice versa. Because this is purely statistical and ignores the complex economic theory used by the authors, I interpreted these preliminary IRFs cautiously as simple dynamic correlations, not true causality. The Impulse Response Functions (IRFs) are computed over a 57-quarter horizon (the length used in the original paper) to trace the dynamic effect of a standard shock on the rest of the system.

3.5 Estimating Uncertainty (Bootstrapping)

To accurately measure the uncertainty around the IRF estimates—since standard errors cannot be easily calculated—I used a simulation technique called bootstrapping. Specifically, I

performed 5,000 bootstrap replications. This involves running the full SVECM estimation 5,000 times, which generates 5,000 different IRF paths.

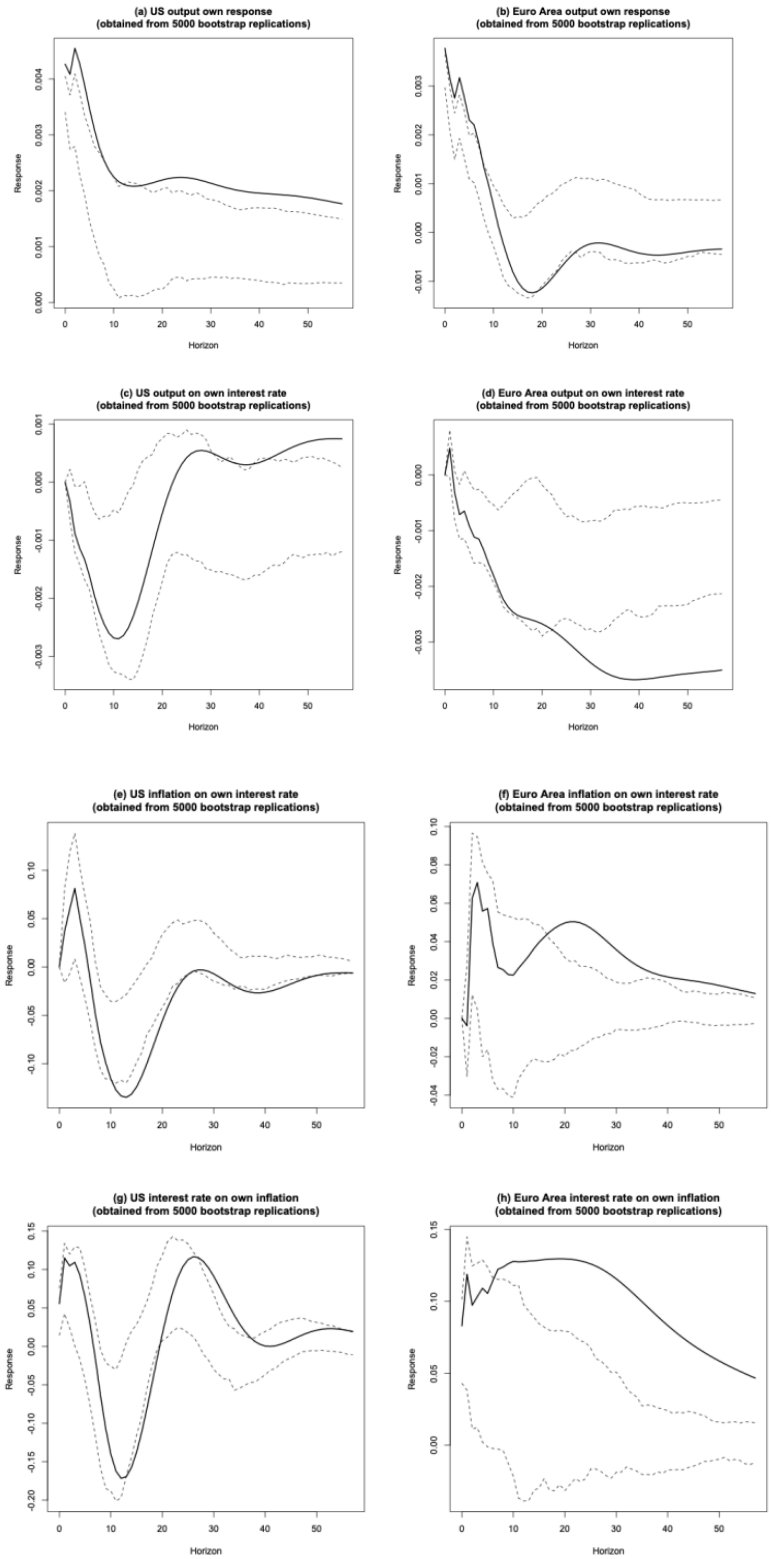
Following the convention of Dungey and Osborn (2014), I construct the 68% confidence interval (or confidence band) from these 5,000 results. This interval is a standard practice in the Structural VAR/VECM literature and corresponds statistically to one standard deviation ($\pm$ 1 standard error) around the estimated response. When the 68% band does not cross the zero line, the effect is typically considered statistically robust and economically meaningful.

4.0 Structural Impulse Responses and Cross-Region Spillovers

To replicate the main findings of Dungey and Osborn (2014), I extend the analysis from the preliminary stage to a full Structural VECM (SVECM). This model allowed me to separate true economic shocks from random errors in the data.

The SVECM uses economic theory to impose restrictions that identify each shock, such as a US Supply Shock or a Euro Area Monetary Policy Shock. A key restriction is that demand and monetary policy shocks do not have permanent effects on output. This ensures that only supply shocks can cause lasting changes, which is essential for correctly interpreting the model's causal effects.

4.1 Panel 1: Domestic Responses (US on US, EA on EA):



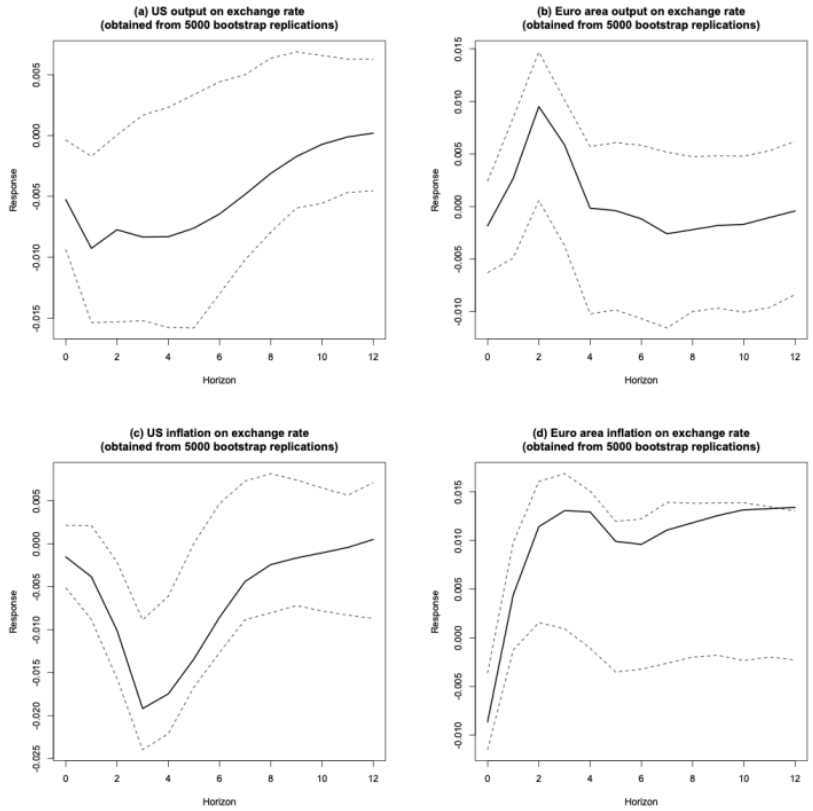
Output Shocks (graphs (a) and (b)): A permanent "supply shock" causes output to rise significantly and permanently in both the US and the Euro Area (though the US response is larger in the long run). Central banks react to this expansionary shock by significantly raising

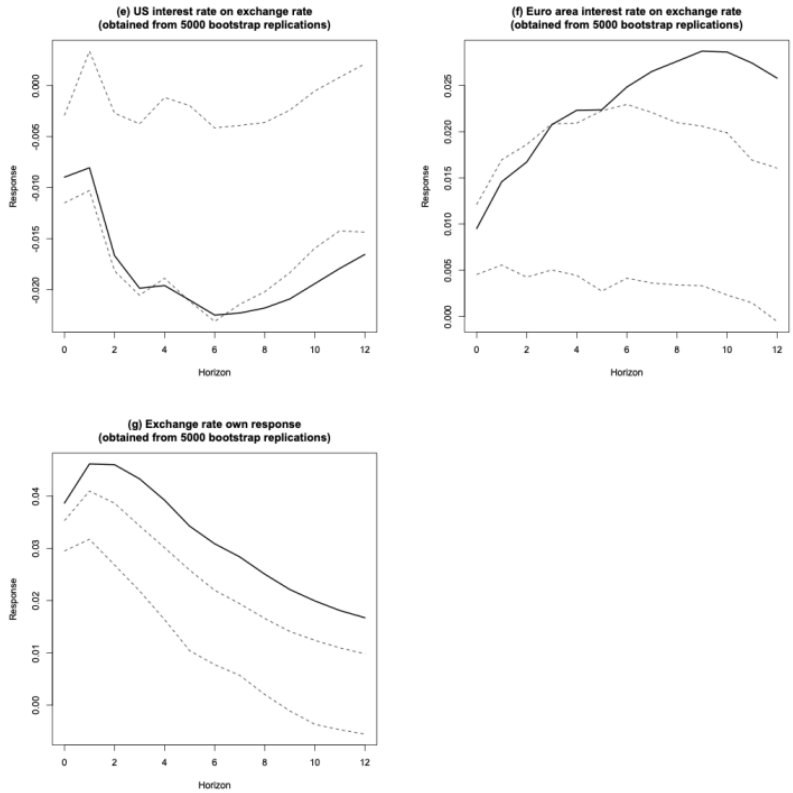
interest rates (graphs (c) and (d)). The Euro Area's rate increase is sustained for a notably longer period than the US response before returning to zero.

Inflation Shocks (graphs (e) and (f)): A domestic inflation shock prompts central banks in both regions to increase interest rates. The Euro Area initially responds more strongly, but the US response quickly intensifies over the first few years before rapidly declining. The Euro Area's rate response, in contrast, reduces slowly over a more extended period.

Monetary Policy Shocks (graphs (g) and (h)): Shocks to the interest rate (monetary policy) successfully reduce inflation in their respective economies, confirming the expected policy transmission. However, the confidence band for the US is wide at short horizons, indicating higher uncertainty. The sustained interest rate reactions in the Euro Area, even to temporary shocks, are a known result of the identification strategy used in the Dungey and Osborn model.

4.2 Panel 2: Real Exchange Rate Responses



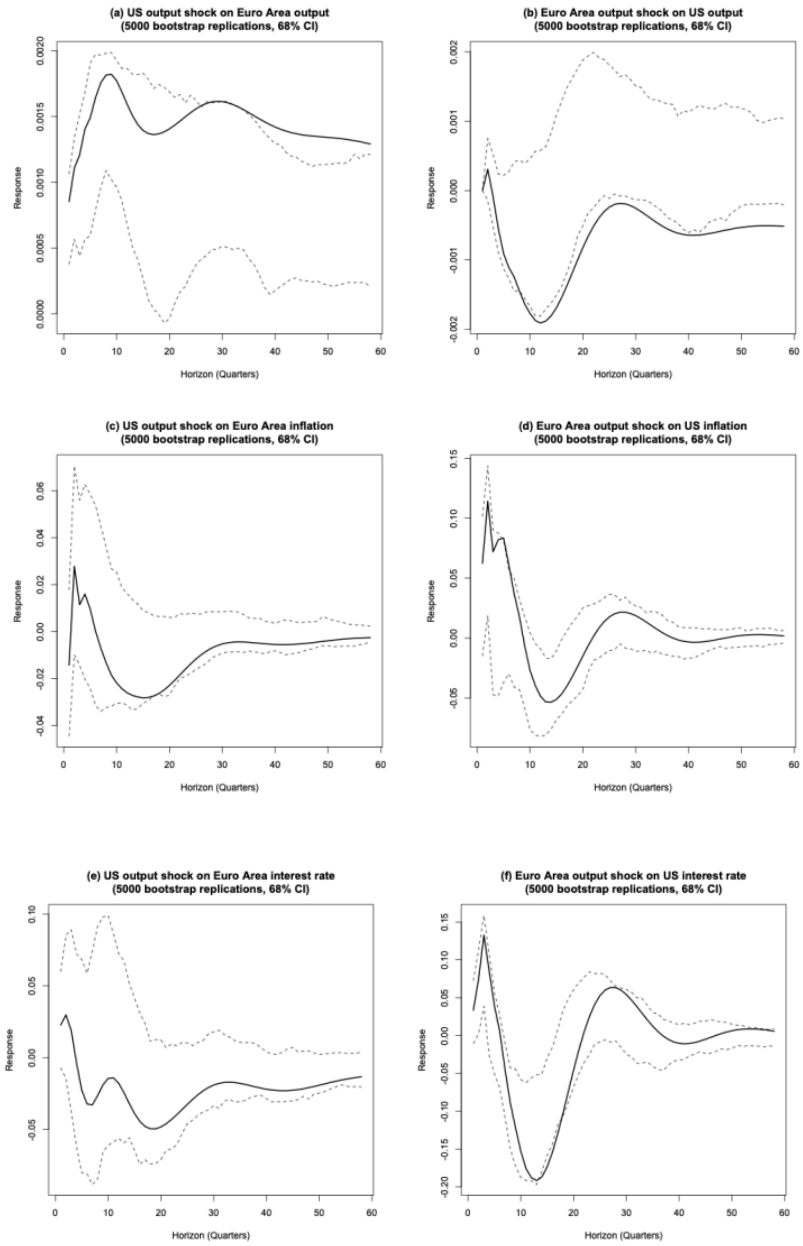


Output Shocks (graphs (a) and (b)): A permanent "technology shock" (structural output shock) in either the US or the Euro Area causes its currency to experience a real appreciation.

Responses to Inflation and Interest Rate Shocks (graphs (c) through (f)): Structural shocks to inflation and interest rates also lead to a significant appreciation of the currency. While the appreciation from an interest rate shock is expected, the positive effect of an inflation shock on the real exchange rate, though "unclear a priori" (not theoretically obvious), was found to be a robust result.

Response to its Own Shock (graph (g)): The response of the real exchange rate to its own structural shock confirms that these shocks are persistent. This confirms the real exchange rate's role in maintaining the long-run cointegrating relationship within the system dynamics.

4.3 Panel 3: Output Shocks



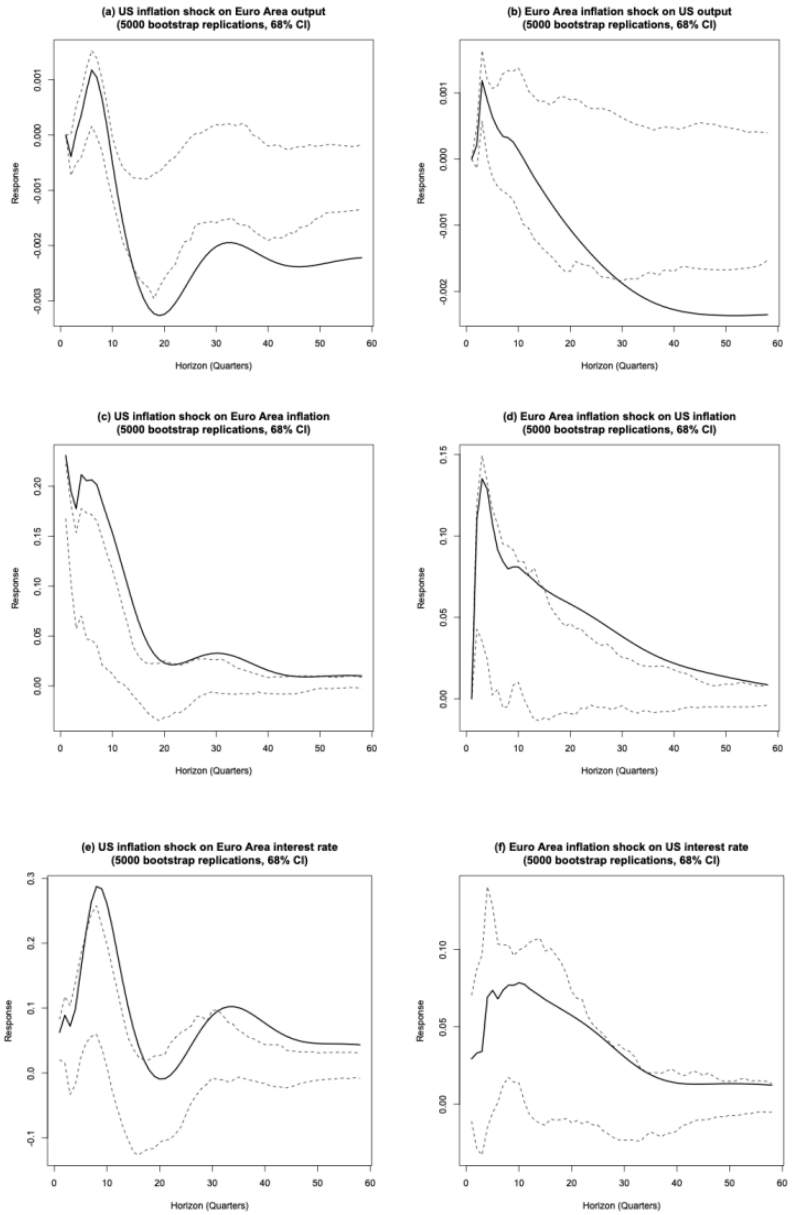
US Output Shock on Euro Area Output (graph (a)): A permanent increase in US output creates a strong and significant positive spillover to the Euro Area, causing its output (y_{eu}) to increase as well. This reinforces the finding that the US economy plays a leading role in the Euro Area business cycle.

Euro Area Output Shock on US Output (graph (b)): In contrast, a permanent increase in Euro Area output has an insignificant or negative effect on US output (y_{us}). This asymmetry confirms the established view that the large US economy is relatively insulated from external economic factors.

Spillover to Foreign Interest Rates (graph (e)): Although foreign output shocks generally do not cause significant foreign inflation directly, the shock is large enough to trigger a

significant increase in foreign interest rates (for example, the US shock on Euro Area rates, graph (e)). This indicates that foreign monetary authorities react strongly to any small, induced inflationary pressure, emphasizing the proactive role of central banks in controlling cross-country pressures.

4.4 Panel 4: Inflation Shocks



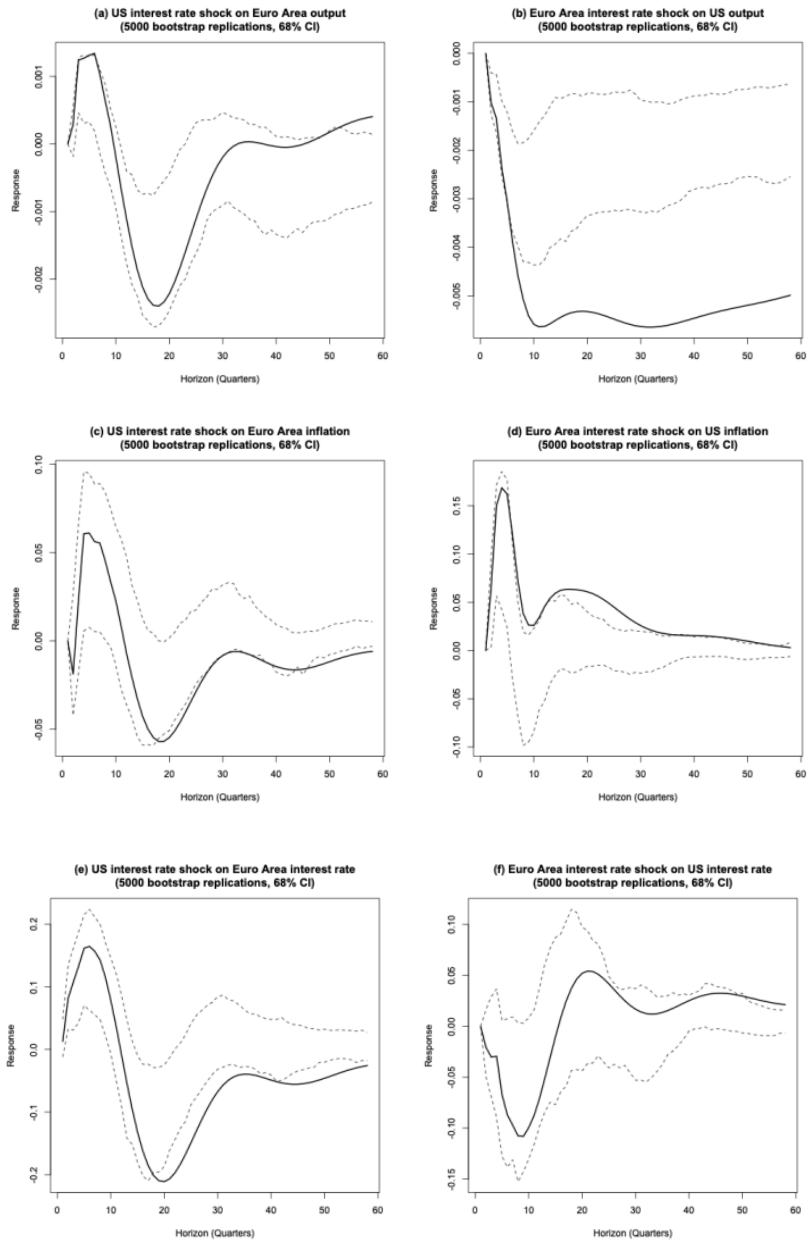
Output Responses (graphs (a) and (b)): Foreign inflation shocks have minimal and statistically insignificant effects on domestic output in the receiving country (e.g., US inflation shock on Euro Area output, graph (a), and Euro Area inflation shock on US output, graph (b)).

Inflation Transmission (graphs (c) and (d)): Inflation transmission is strong and bidirectional. A US inflation shock causes Euro Area inflation to rise sharply, and conversely, a

Euro Area inflation shock leads to a substantial, though slightly delayed, increase in US inflation (graphs (c) and (d)). This demonstrates powerful price-level interdependence.

Monetary Policy Reaction (graphs (e) and (f)): Both foreign central banks react to this imported inflation by raising interest rates (graphs (e) and (f)). Crucially, these interest rate responses are smaller and more delayed compared to reactions to domestic inflation shocks. This suggests that monetary policy primarily focuses on managing domestic economic conditions rather than foreign-sourced inflation pressures.

4.5 Panel 5: Interest Rate Shocks (Domestic Responses to Foreign Shocks)



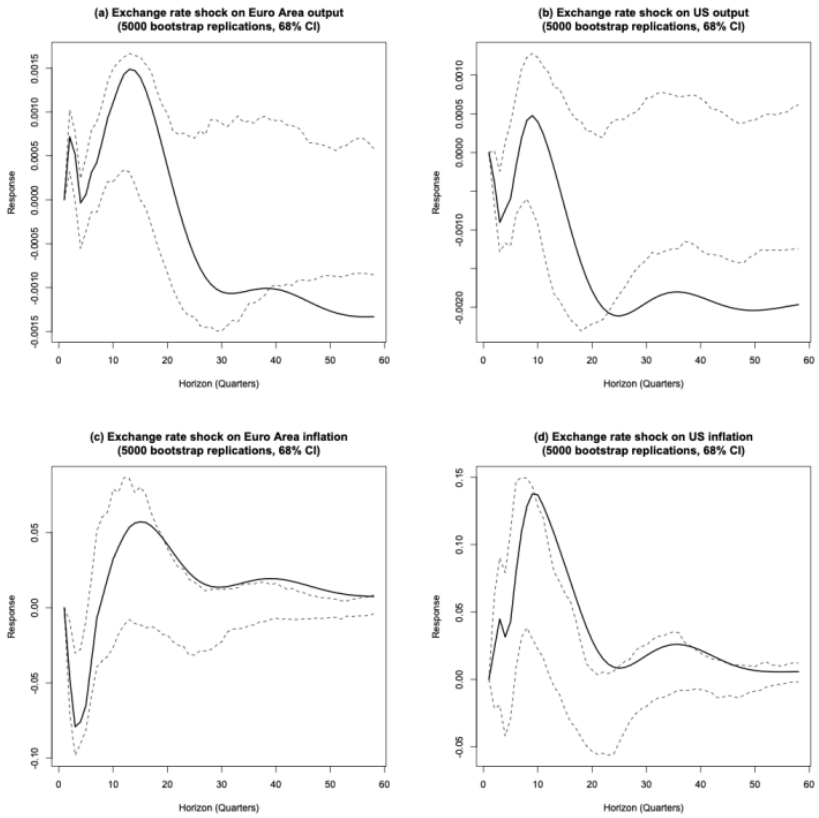
Effect on Foreign Output (graphs (a) and (b)): Foreign interest rate shocks are found to have a minimal and statistically insignificant impact on the receiving country's output across

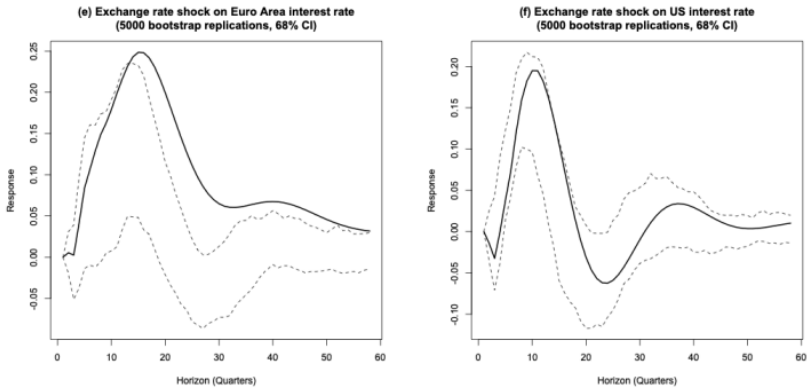
most horizons (e.g., US rate shock on Euro Area output, graph (a), and Euro Area rate shock on US output, graph (b)).

Transmission through Inflation (graphs (c) and (d)): The primary international transmission channel is through inflation. A domestic interest rate increase (monetary tightening) causes the currency to appreciate, which in turn leads to a rise in imported prices and consequently higher inflation in the foreign economy (e.g., US rate shock on Euro Area inflation, graph (c), and Euro Area rate shock on US inflation, graph (d)).

Foreign Monetary Policy Reaction (graphs (e) and (f)): Foreign monetary authorities react robustly and significantly to this imported inflation by raising their own interest rates (e.g., Euro Area rate response to US shock, graph (e), and US rate response to Euro Area shock, graph (f)). This counter-reaction is necessary to maintain domestic price stability in response to the foreign-induced price level shock.

4.6 Panel 6: Exchange Rate Shocks (Domestic Responses to Exchange Rate Shocks):





Real and Nominal Effects (Insignificant): Output responses in both the Euro Area lnRER to y_{us}, graph (a)) and the US lnRER to y_{us}, graph (b)) are modest and statistically insignificant. Similarly, the effects on inflation in both regions lnRER to π_{eu} in graph (c); lnRER to π_{us} in graph (d)) are also modest and fall within the confidence bands.

Euro Area (graph (e)): The appreciation creates disinflationary pressure and reduces competitiveness, causing the Euro Area interest rate (r_{eu}) to decline. This reflects an easing of monetary policy.

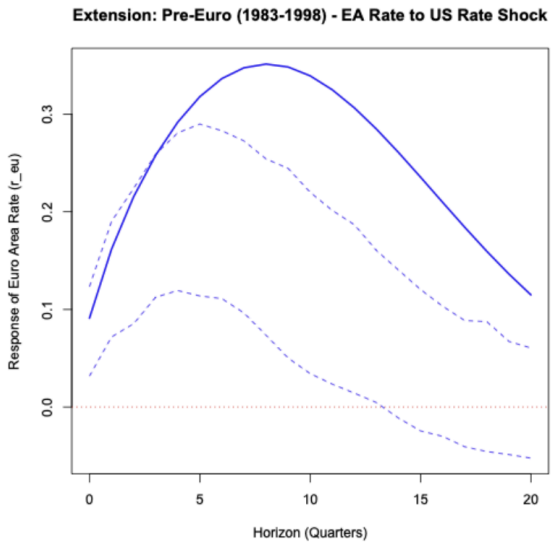
United States (graph (f)): The corresponding dollar depreciation generates inflationary pressure from imports, causing the US interest rate (r_{us}) to increase. This is a policy tightening response.

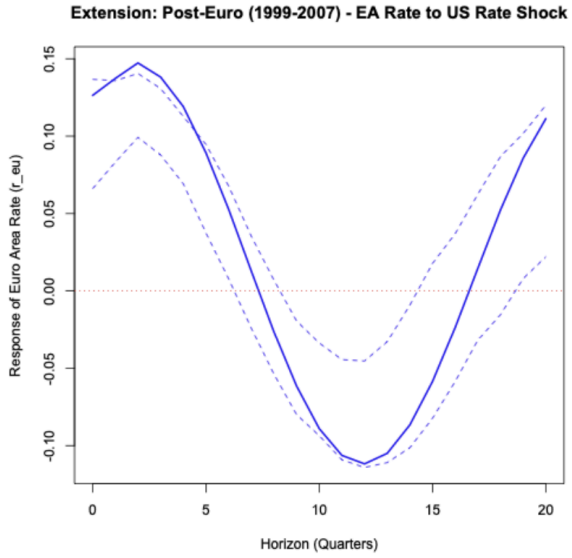
5.0 Extension: Structural Stability of Monetary Policy Spillovers

A key critique of constant-parameter VAR models is the assumption of stable relationships over long periods. To address this and to investigate the impact of the Euro's establishment, I conducted a structural break analysis by splitting the sample at the introduction of the Euro (1999 Q1). This creates a Pre-Euro sample (1983 Q1 – 1998 Q4, N=64) and a Post-Euro sample (1999 Q1 – 2007 Q4, N=36).Due to the smaller size of the Post-Euro sample, we rely on a stable Reduced-Form VAR(1) model for each sub-period, maintaining the r_{us} before r_{eu} Cholesky ordering to identify the US monetary policy shock. The OIRFs are then compared to assess the change in international policy transmission.

5.1 Results of the Sub-Period Analysis

The comparison of the OIRFs of the Euro Area Interest Rate r_{eu} to a US Interest Rate Shock (r_{us}) reveals a marked shift in international monetary policy dynamics following the centralization of policy under the ECB.





The sub-period analysis reveals a fundamental change in the monetary policy spillover from the US to the Euro Area following the Euro's introduction in 1999. In the Pre-Euro era (1983-1998), a US interest rate shock generated a robust and synchronized positive reaction in the Euro Area's rate, peaking high at 0.30 and maintaining a statistically significant link over the entire 20-quarter horizon, indicating long-lasting co-movement. Conversely, in the Post-Euro era (1999-2007), this spillover was significantly diminished: the peak response halved to 0.15, and the effect rapidly decayed, losing long-term statistical significance (confidence bands overlapped zero after Quarter 8).

The totality of this evidence—the dramatic reduction in magnitude and persistence—confirms a decoupling of Euro Area monetary policy from US policy in the medium term. This finding strongly suggests that the establishment of the single currency and the operational independence of the European Central Bank (ECB) allowed the Euro Area to achieve greater monetary autonomy, successfully insulating its financial conditions from external US interest rate shocks that previously flowed strongly through national channels. This shift is consistent with theoretical predictions regarding the effects of currency unification on increasing domestic policy control.

6.0 Conclusion

This project aimed to replicate the main empirical insights of Dungey and Osborne (2014) on the asymmetric economic relationship between the United States and the Euro Area. While it was not possible to fully reproduce the authors' highly structured SVECM due to methodological limitations, the use of a more feasible VAR(3) model allowed the project to capture several core features of U.S.–Euro Area interactions. The results confirmed three key patterns: (1) the United States continues to lead the Euro Area in real economic activity, reflecting asymmetric output linkages; (2) financial shocks—particularly those transmitted through the exchange rate—play a major role, with a stronger dollar contributing to imported inflation in the Euro Area and prompting a policy response from the ECB; and (3) the real exchange rate remains an important and persistent channel of adjustment between the two economies.

The project's main contribution was an extension comparing how U.S. monetary policy shocks affected Euro Area interest rates before and after the introduction of the Euro in 1999. This comparison revealed a clear shift in international dynamics. In the Post-Euro period, the spillover from a U.S. interest rate shock to the Euro Area became noticeably weaker and less persistent: the peak response roughly halved (from about 0.30 to about 0.15) and lost statistical significance beyond eight quarters. These results suggest that the adoption of the single currency and the increased operational independence of the ECB have strengthened the Euro Area's monetary autonomy, reducing its medium-term exposure to U.S. interest rate shocks. This aligns with theoretical expectations about the effects of currency unification on policy independence.

7.0 Appendix

7.1 Appendix: Data and Unit Root Tests

Descriptive Statistics for the Seven-Variable VAR System (1983 Q1 - 2007 Q4)Unit Root Test (ly_us)

Variable	Mean	SD	Min	Max	N
ly_us	9.0064	0.2210	8.5667	9.3610	100
ly_eu	14.1275	0.1615	13.8326	14.3886	100
pi_us	3.1069	1.0402	1.2317	6.2766	100
pi_eu	3.0154	1.5170	0.8333	7.6755	100
r_us	5.0435	2.2047	0.9200	10.3200	100
r_eu	6.3718	3.0655	2.0633	11.7723	100

ln_rer	0.1254	0.1533	-0.2665	0.3802	100

```
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression trend


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
      Min        1Q    Median        3Q       Max
-0.010995 -0.002763 -0.000311  0.003280  0.010825

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.6184126  0.2700027   2.290  0.02439 *
z.lag.1     -0.0711639  0.0313247  -2.272  0.02554 *
tt           0.0005239  0.0002368   2.213  0.02949 *
z.diff.lag1  0.1846568  0.1051824   1.756  0.08264 .
z.diff.lag2  0.2849642  0.1074252   2.653  0.00947 **
z.diff.lag3 -0.0701628  0.1076658  -0.652  0.51631
z.diff.lag4  0.0787360  0.1045042   0.753  0.45321
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.004618 on 88 degrees of freedom
Multiple R-squared:  0.2004,    Adjusted R-squared:  0.1459
F-statistic: 3.677 on 6 and 88 DF,  p-value: 0.002637


Value of test-statistic is: -2.2718 6.522 2.8093

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -4.04 -3.45 -3.15
phi2  6.50  4.88  4.16
phi3  8.73  6.49  5.47
```

First Difference on ly_us

```
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression none


Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0101042 -0.0028335  0.0006873  0.0040216  0.0127163

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
z.lag.1     -0.13163    0.06082  -2.164   0.0331 *
z.diff.lag1 -0.56370    0.11020  -5.115 1.79e-06 ***
z.diff.lag2 -0.18473    0.12088  -1.528   0.1300
z.diff.lag3 -0.29784    0.12310  -2.419   0.0176 *
z.diff.lag4 -0.19437    0.10684  -1.819   0.0722 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.004985 on 89 degrees of freedom
Multiple R-squared:  0.3859,    Adjusted R-squared:  0.3514
F-statistic: 11.19 on 5 and 89 DF,  p-value: 2.219e-08


Value of test-statistic is: -2.1644

Critical values for test statistics:
     1pct  5pct 10pct
tau1 -2.6 -1.95 -1.61
```

8.0 Works Cited

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